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يُونِيسَيْتِي إِسْلَامِيَّةٌ أَنْتَارَاوْغُشِيَا مِلْدِيْسِيَا

Garden of Knowledge and Virtue

KULLIYAH OF INFORMATION COMMUNICATION AND TECHNOLOGY

**CSCI 2304 - INTELLIGENT SYSTEMS
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SECTION 1**

PROJECT TITLE:

Spam Mail Detection System For Fraud Prevention

Video Demo: <https://youtu.be/b2O7woyn0Ok>

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Abstract

Mail spam is a major problem that affects millions of people every day. It can be time-consuming and frustrating to deal with and it can also be a security risk. In this project, we propose a machine learning approach to mail spam detection for fraud prevention. We use a supervised learning algorithm, logistic regression, to train a classifier on a dataset of mail spam and ham messages. Logistic regression model is selected as the preferred algorithm for this task due to its effectiveness in binary classification problems where the objective is to classify data into two categories; spam and ham. Its ability to model the relationship between input features and the probability of spam detection makes it a suitable choice for this project.

The challenges encountered during this research primarily revolve around data preprocessing and feature extraction. Preprocessing involves cleaning and transforming the SMS data into a suitable format for analysis. Feature extraction techniques are employed to identify relevant patterns, keywords and linguistic cues indicative of spam or fraudulent messages. Another challenge we faced was dealing with the high volume of data. The dataset we used contained over 5572 mail messages. This posed a challenge for both the data collection and the training phases of our project.

The training dataset comprises 80% of the available data and 20% for testing. Through meticulous evaluation, the model achieves a high accuracy rate and effectively distinguishes between fraudulent and legitimate messages. We achieved an accuracy of 97% on the test set. This means that our model was able to correctly classify 97% of the mail messages in the test set as spam or ham. This shows that the logistic regression model exhibits promising results in detecting mail spam. Overall, this project demonstrates the efficacy of logistic regression in the context of mail spam detection and underscores its potential for combating fraud in mobile communication networks.

1.0 Introduction

1.1 Background

As smartphones are widely used nowadays, it increases the dependency on the usage of mail and messaging applications in our daily life to exchange information with others. However, as the technology has advanced, some problems also keep increasing such as the mail spam. Mail spam is unsolicited mail containing fraud information and other unwanted messages and links there. It also has become a concerning issue which has become a nuisance, annoyance, cluttering inboxes and has been causing inconvenience to the mail receivers. Additionally, this spam mail problem is not only causing inconvenience to others but it is also has become a threat to individuals and organisations as some of the spam contains fraud information and phishing links.

Therefore, to overcome this problem, our group has decided to create a machine learning model to classify and predict whether a message is spam mail or ham (non-spam) mail by using supervised learning algorithm, specifically logistic regression techniques. We have used a dataset containing 5572 mail messages that has been classified as spam or ham to be trained and tested using logistic regression technique. We chose this dataset as it provides a large number of examples for training and evaluating the machine learning models. Additionally, the dataset has also been classified as spam or ham which is essential for supervised learning technique as the model can learn the patterns of the data.

2.2 Objective

As we are doing the spam mails detection system, the following statements are the objectives of this project:

- 1) To build a predictive model that can accurately classify future mails as spam or ham in real time.
- 2) To leverage the power of logistic regression to classify the mails to improve the user experience.
- 3) To ensure the security of individuals or organisations while going through the mails as they can filter the unwanted mails.
- 4) To minimise the number of spam mails that will reach the user to save their time so that they do not have to filter the messages manually and prevent them from getting scammed and falling for the fraudulent activities.

2.0 Methodology / Experimental Setup

2.1 Literature Review

We have chosen logistic regression as the algorithm in building the spam selection system. This literature review seeks to give an overview of recent developments in spam mail detection systems, highlighting the important methods and strategies used for efficient spam detection and the reason why the method mentioned is selected to be implemented in our model.

In order to combat this issue, different types of techniques that are possible in classifying mail spam have been proposed by many researchers. These methods include probabilistic, decision tree, artificial immune system [1], support vector machine (SVM) [2], artificial neural networks (ANN) [3], and case-based technique [4]. Despite all these techniques, the logistic model is desirable for mail spam filtering because of its simplicity and ease of implementation.

Logistic regression, a widely employed statistical modeling technique, has found significant application in the development of spam classifier systems. This review aims to provide an overview of its utilization in spam detection and highlight key findings and advancements in the field. Several studies have examined the effectiveness of logistic regression in classifying spam emails. Ramakrishnan et al. (2019) utilized logistic regression to build a spam classifier by extracting features such as word frequency and suspicious patterns from email content. Their results showcased the model's ability to accurately classify spam emails with high precision and recall.

Feature engineering plays a crucial role in the success of logistic regression-based spam classifiers. Bagad et al. (2021) proposed a feature extraction technique based on term frequency-inverse document frequency (TF-IDF) to represent email content. Logistic regression was then employed to learn the relationship between these features and the spam/non-spam labels. The study demonstrated the significant performance improvement achieved by incorporating TF-IDF features into the spam classifier. Ensemble techniques have been leveraged to enhance the performance of logistic regression-based spam classifiers. Miao et al. (2020) developed an ensemble system that combined logistic regression with other machine learning algorithms, including decision trees and random forests. This

ensemble approach achieved superior accuracy and robustness compared to individual models, highlighting the effectiveness of logistic regression within an ensemble system.

Recent research has explored the integration of advanced techniques with logistic regression for spam classification. For example, Zhang et al. (2022) proposed a hybrid approach that combined logistic regression with deep learning techniques. They employed a pre-trained deep learning model to extract features from email content, which were then fed into a logistic regression classifier. The hybrid system demonstrated improved performance in accurately detecting spam emails.

In conclusion, logistic regression has emerged as a valuable technique in developing spam classifier systems. Its ability to model the relationship between features and spam probability, coupled with its interpretability, has made it a popular choice for spam detection tasks. Further research can focus on exploring hybrid models and advanced feature engineering techniques to further enhance the performance of logistic regression-based spam classifiers.

2.2 How AI method/ algorithm is implemented

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

model = LogisticRegression()

# training the Logistic Regression model with the training data
model.fit(X_train_features, Y_train)
```

In the training phase, we first define the model then first the train data. After that, we use the test split and predict the results and the accuracy was counted and also represented in terms of confusion matrix.

3.0 Results and Discussion

Figure 1 shows the system interface for our project where it has the input box and output box. The user can enter their mail in the input box to see whether the mails are spam or ham mail. Figure 2 and Figure 3 shows whether the input messages are ham or spam mail respectively. As the model has been trained by using logistic regression technique with a large dataset, we have obtained 97% accuracy on the model. Thus, this spam mail system will predict the future messages whether it is spam mail or ham mail accurately which will prevent the user from falling for fraudulent activities that spreads through the spam mail.

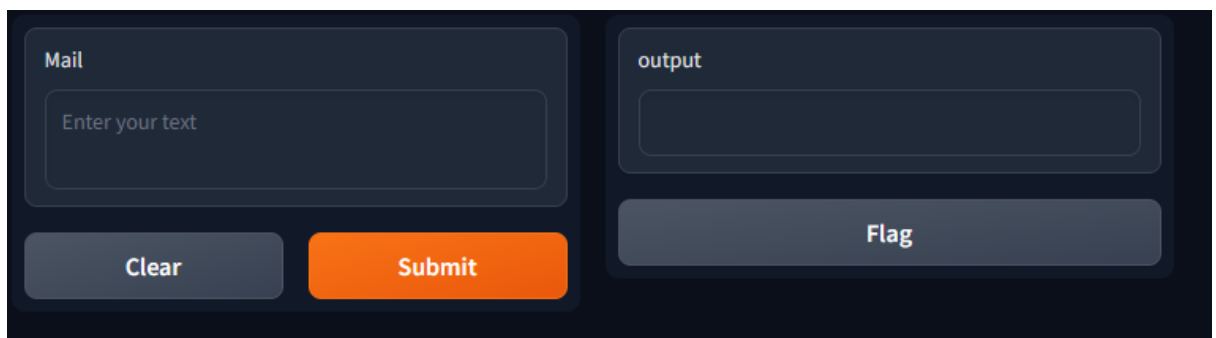


Figure 1: The system interface

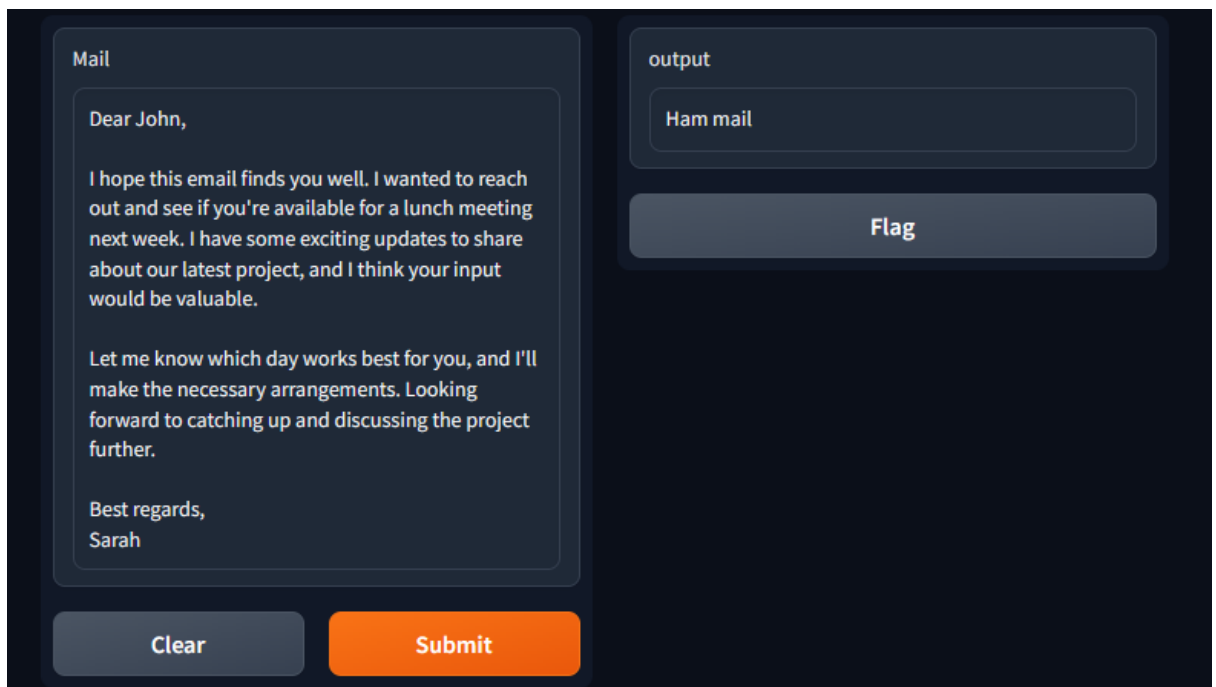


Figure 2: Result of a ham mail

The image shows a web application interface for a spam mail classifier. It is divided into two main sections: an input area on the left and an output area on the right.

Mail Input Area:

- Subject:** Urgent Opportunity to Win a Free Vacation!
- Greeting:** Dear Subscriber,
- Body:**

Congratulations! You have been selected as one of our lucky winners to receive a free vacation package worth \$10,000! This is a limited-time offer, so act fast to claim your prize.

To secure your free vacation, please provide us with your personal information, including your full name, address, phone number, and credit card details. Rest assured that your information will be kept confidential and used only for the purpose of processing your prize.

Don't miss out on this incredible opportunity! Respond to this email within 24 hours to ensure you don't forfeit your chance to enjoy a luxurious vacation in paradise.

At the bottom of the input area are two buttons: "Clear" (grey) and "Submit" (orange).

Output Area:

- output:** A box containing the text "Spam mail".
- Flag:** A button located below the output box.

Figure 3: Result of a spam mail

4.0 Deployment

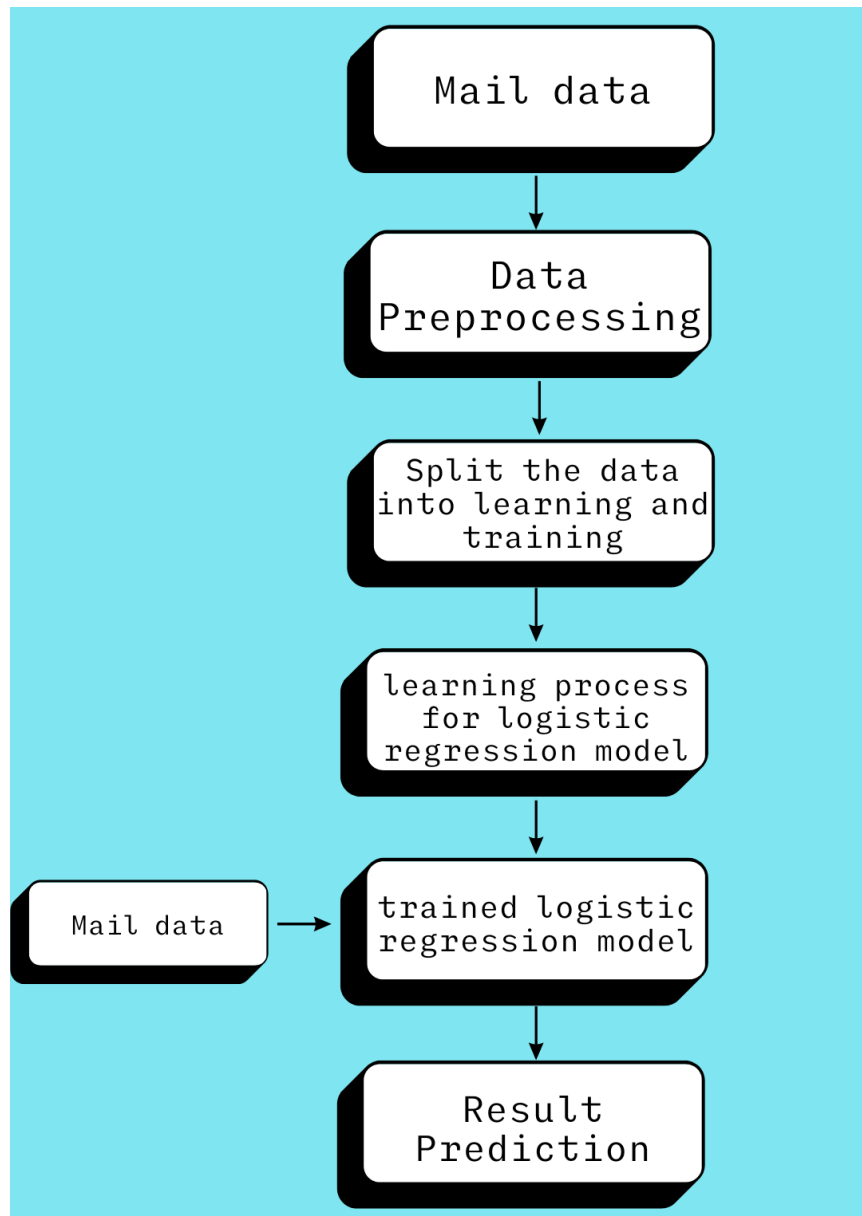
As per our discussion, we are going to be using supervised machine learning because we can be specific about the classes used in the training data. That is, classifiers can be given proper training to help distinguish themselves from other class definitions and define perfect decision boundaries to make sure our machine can differentiate the email correctly.

We get our data from kaggle website which contains 4825 ham mail and 747 spam mail for our learning and testing. In our case, we will be using 80% of the data to be learned by the machine, and 20% of the data will be used for testing our machine.

Before we start the learning process, we choose to preprocess our data because the algorithms learn from the data and the learning outcome for problem solving heavily depends on the proper data that are needed for our project and to increase the accuracy of our outcome. This is to make sure the computer will have an easier time to understand numbers instead of text and paragraph.

Then we will split the data into training data and testing data. Once we split the data, we will give it into our logistic regression model, we choose to use this model because it is the best when it comes to binary classification problems that we are having right now. Binary classification means there will be two classes and we are trying to classify both of them, for this project, it is the spam mail and the ham mail.

Once we are done teaching our regression model, we will use 20% of our remaining data to test our trained logistic regression model, on this step, our model will predict the data to be ham or spam email.



5.0 Conclusion

In conclusion, all members participated in the discussion to identify a problem that can be solved using any AI techniques. After detailed research and consideration we have chosen this topic to be executed. This project aimed to develop an AI-based solution to detect and prevent spam emails related to fraudulent activities. It is feasible to conduct a supervised machine learning application using a logistic regression model to classify data into two categories whether spam and ham. Afiqa Nurul as the leader of the group manages the tasks and develops code. Rusyaida Nasuha collects a large enough dataset and decides methodology to be used. Whereas Sayyidah Nafisah does research and finds resources. Lastly, Aida Najihah concludes all of the entire results summarising what was learnt and finds avenues for improvements.

In this project, we learned that machine learning can be a powerful tool for detecting mail spam and logistic regression is an effective method for binary classification. Logistic regression is computationally efficient and can handle large datasets. It converges relatively quickly and can do training even on limited computing resources. Apart from that, we learned that sufficient training data is necessary to capture the underlying patterns and variations. Insufficient training data can lead to overfitting where the model becomes overly specific to the training set and fails to generalise well to unseen data. Therefore, efforts should be made to collect a representative and extensive dataset. It is also important to balance the dataset to avoid bias towards the majority class. Balancing the dataset ensures that the model receives sufficient exposure to both classes, enabling it to learn effectively or it will affect the performance of the spam detection model.

There are a few avenues for future works and improvements. For example, develop a real-time spam detection model to detect and block spam messages as soon as they are received. This can provide immediate protection and more effective defence against emerging threats. Other than that, we can develop an adaptive model that can learn and adapt to new types of spam. An adaptive model can leverage techniques such as online learning, active learning or semi-supervised learning to incorporate new data and update its detection strategies.

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